Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? The answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only σ-additivity, and the choice of state space is unconstrained by the algebra [18].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism splits into two grades: Gleason-type results supply a σ-additive measure on the *lattice* ($\mathrm{PR}_{\mathrm{lat}}$), but the passage to a measure on the *dual space* ($\mathrm{PR}_{\mathrm{dual}}$) fails — for the central case $L(H)$, no normal state extends to a charge on the dual, so even probabilistic realism in the dual-space sense is obstructed, and value-definite realism fails outright.

Distributivity guarantees value-definiteness; for the physically central case $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. The state space admits a natural filtration $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\mathrm{df}}$, whose three transitions — pasting, regularity, sharpness — are of different mathematical character. Distributivity collapses the filtration; non-distributivity can (but need not) open it.

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1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [22] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation.

Three nested positions, each a strengthening of the previous:

(i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.

(ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

(iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value: a global dispersion-free state on the full observation algebra. On a Boolean algebra (and on $L(H)$) this coincides with a 2-valued homomorphism; on a general OML the two come apart (Remark below), and it is the dispersion-free state that is the operative notion throughout.

Both EA and PR as stated presuppose a genuine measure on the dual space: EA *is* such a measure, and PR descends from it. In the Boolean case this presupposition is free (the Stone construction supplies the dual-space measure unconditionally). In the OML case it is not, and PR accordingly refines into two grades — $\mathrm{PR}_{\mathrm{lat}}$, a measure on the lattice, and $\mathrm{PR}_{\mathrm{dual}}$, a measure on the dual space concentrating on the physical points — which coincide for Boolean algebras and separate for $L(H)$ (§3.1). Where the distinction matters we write the grade explicitly; an unqualified "PR" refers to the Boolean case, where the two agree.

These are nested: VDR implies PR implies EA. Which implications reverse depends on the observation algebra:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn's lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ-additivity of the charges [18]; the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors[1] — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [17] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism on the *lattice* remains available via Gleason's theorem [9], but its *dual-space* form fails for normal states (no extension to a charge on $S_0(L(H))$); empirical adequacy is unconditional in the $\perp$-stable set-function sense.

Distributivity guarantees value-definiteness; for $L(H)$ with $\dim H \geq 3$, its absence blocks it. The individual ingredients — Stone duality, the Kochen–Specker theorem, Gleason's theorem, the Bub–Clifton uniqueness result, and the recent McDonald–Bimbó duality for orthomodular lattices [19] — are known. What has not been stated is the synthesis: that distributivity of the observation algebra is the condition controlling which of the three positions are available, and that the state space admits a filtration whose transitions are of different mathematical character. We begin with the Boolean case [18].

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i\in\iota}$ with injective homomorphisms $\varphi_{ij} : B_i \to B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra*.

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with each $\ell_i : B_i \to [0, 1]$ finitely additive, $\ell_i(1) = 1$, and

$$\ell_j \circ \varphi_{ij} = \ell_i \qquad (i \leq j).$$

The Stone space $\mathrm{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter

$$\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}.$$

[1] The non-quantum examples require the incompatibility to be *operationally imposed*: the contexts must be genuinely non-co-realizable, with no joint probability space carrying all observations at once. This is the content of contextuality as failure of a global section (Abramsky–Brandenburger [1]; Fine's theorem [8] for the joint-distribution criterion). Incompatibility cannot arise *intrinsically* from the dynamics of a single classical system: any finite family of classical observables on a common probability space jointly distributes (Kolmogorov), hence sits inside a Boolean algebra. Distributed-sensor contextuality is therefore a statement about non-co-realizable measurement settings, not about the dynamics of one underlying process.

Any compatible family of normalised charges extends to a σ-additive Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ — unconditionally, with no assumption of σ-additivity [3, Theorem 1], [18]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

The measure $\hat{\mu}$ lives on $\mathrm{St}(\mathcal{C})$, not on Ω. Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires

$$\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1,$$

i.e., that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each ℓ_i is σ-additive [18].

Every Boolean algebra admits lattice ultrafilters — maximal proper filters — by Zorn's lemma. On a Boolean algebra, maximality forces a dichotomy: for every a, either $a \in F$ or $a^c \in F$. The proof extends F to include a^c whenever $a \notin F$; the extension remains proper because $a^c \wedge f \neq \perp$ for all $f \in F$, which follows from distributivity. This dichotomy makes every lattice ultrafilter a 2-valued homomorphism, assigning each $a \in \mathcal{C}$ a definite value in $\{0, 1\}$. On a non-distributive OML, the extension argument fails, and maximal proper filters need not be 2-valued.

Moreover, any subset $S \subseteq \mathrm{St}(\mathcal{C})$ that projects surjectively onto each $\mathrm{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω [18].

Proposition 1.1 (Boolean commensurability)**.** *For a directed system of Boolean algebras with compatible charges:*

(a) EA is unconditional: $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ always exists.

(b) PR holds iff each ℓ_i is σ-additive.

(c) VDR holds unconditionally: $\mathrm{St}(\mathcal{C})$ consists entirely of 2-valued homomorphisms.

The only obstruction is σ-additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.

Proof. Part (a) is the Stone construction [3, 18]. Part (b) is [18, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. □

2 Orthomodular observation algebras

Proposition 1.1 depends on distributivity at exactly one point: the existence of 2-valued homomorphisms. When observations are incompatible, this fails. The non-distributive structure need not be postulated: Gunji et al. [10] show that gluing Boolean algebras along shared boundaries via categorical pushout produces an orthomodular lattice, generically non-distributive. The OML is *forced* by the context structure.

Definition 2.1. An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^\perp, 0, 1)$ with an orthocomplementation ${}^\perp : A \to A$ satisfying $a \wedge a^\perp = 0, \quad a \vee a^\perp = 1, \quad a^{\perp\perp} = a$, and $\quad a \leq b \Rightarrow b^\perp \leq a^\perp$.

Definition 2.2. An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [19, Proposition 2.3].

Example 2.3. The lattice $L(H)$ of closed linear subspaces of a separable Hilbert space H with $\dim H \geq 3$, ordered by inclusion with $U^\perp = \{x \in H : \langle x, y\rangle = 0\ \forall y \in U\}$, is an orthomodular lattice that is not distributive [16, Chapter 1].

Example 2.4 (Incompatible measurements)**.** Three binary measurements A, B, C, where (A, B) and (A, C) can be performed jointly but (B, C) cannot. The jointly measurable pairs generate Boolean subalgebras; the full observation algebra is orthomodular, not Boolean.

Two elements a, b in an OML *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of A.

Definition 2.5. A *state* on an OML A is a function $s : A \to [0, 1]$ with $s(1) = 1$ and

$$a \perp b \quad \Longrightarrow \quad s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^\perp$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

Definition 2.6. A state s is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state assigns a definite truth value to every proposition in A. Every 2-valued homomorphism is dispersion-free; the converse holds on Boolean algebras (where dispersion-free states are exactly the ultrafilters) and on $L(H)$, but *fails* on a general OML: a dispersion-free state need not be multiplicative across incompatible propositions. The minimal witness is MO_2 (the horizontal sum of two Boolean 2^2 blocks): assigning 1 to one atom of each block gives a dispersion-free state s, yet for the gap pair a, b with $a \wedge b = 0$ and $s(a) = s(b) = 1$ one has $s(a \wedge b) = 0 \neq 1 = s(a) \wedge s(b)$, so s is not a homomorphism. Throughout, the operative notion of value-definiteness is the dispersion-free state (equivalently, non-emptiness of $\mathcal{S}_{\mathrm{df}}(A)$), not the strictly stronger 2-valued homomorphism.

Theorem 2.7 (Kochen–Specker [17])**.** *If* $\dim H \geq 3$*, there is no dispersion-free state on* $L(H)$*.*

For $\dim H = 2$, dispersion-free states exist; the obstruction is specific to $\dim H \geq 3$.

Theorem 2.8 (Gleason [9])**.** *If* $\dim H \geq 3$*, every* σ*-additive state on* $L(H)$ *is of the form* $s(P) = \mathrm{tr}(\rho P)$ *for a unique density operator* ρ *on* H*.*

The analogue of Stone duality for OMLs is the McDonald–Bimbó duality [19].

Let A be an OML. A *filter* on A is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by a is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on A, and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

Definition 2.9 (McDonald–Bimbó [19])**.** Let A be an OML. The *dual space* of A is the ordered relational topological space

$$S_0(A) = \big(F(A),\ \subseteq,\ \perp_A,\ P(A),\ \mathcal{T}\big),$$

where:

(i) $F(A)$ is the set of all filters on A;

(ii) $\subseteq$ is set inclusion;

(iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;

(iv) $P(A)$ is the collection of principal filters;

(v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

Theorem 2.10 (McDonald–Bimbó [19]). *$S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p-morphisms. Moreover, $A \cong \mathrm{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$-stable subsets of $S_0(A)$.*

The critical differences from Stone duality:

Feature	Boolean (Stone)	OML (McDonald–Bimbó)
Dual points	Ultrafilters	All filters
Distinguished subset	None	$P(A)$ (principal filters)
Clopens	Boolean algebra	OML
Orthogonality	Trivial	Non-trivial ($\perp_A$)
2-valued homomorphisms	Plentiful (Zorn)	May not exist (KS)

In Stone duality, the principal filters (i.e., the principal ultrafilters pure(ω) for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of Ω [18].

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 2.9) explicitly reference $P(A)$. This is the algebraic formalisation: for non-distributive algebras, the distinction between realised and unrealised observation-patterns is constrained by the algebra. Distributivity is the condition under which this constraint vanishes.

$S_0(A)$ is compact, so the topological machinery applies. But the algebraic input is weaker. A state s on A defines

$$\mu(h(a)) = s(a)$$

on $\mathrm{CO}(S_0(A))^\dagger$.

In the Boolean case, $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ is a Boolean algebra of sets, and s is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\mathrm{St}(\mathcal{C})$ gives σ-subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\mathrm{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state s is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which s is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (Theorem 2.8) supplies a σ-additive measure *on the lattice* $L(H)$ for every σ-additive state: every such state arises from a density operator via the Born rule. This is a measure on A itself, and is *not* the same as a measure on the dual space $S_0(A)$ — a distinction the Boolean case obscures and the OML case forces. Indeed the passage to the dual space *fails* for $L(H)$: no normal (Gleason) state extends to a charge on the clopens of $S_0(L(H))$, by a clustering argument inside a single two-dimensional subspace.[2] For general OMLs the extension problem is open.[3]

[2]Fix a 2-dimensional subspace e. For k distinct lines $p_1, \ldots, p_k \subset e$ the $2k$ lines $\{p_i, p_i^\perp\}$ are pairwise meet-zero, so h sends them to pairwise-disjoint clopens, and a charge forces $\sum_i [s(p_i) + s(p_i^\perp)] = k\, s(e) \leq 1$ for every k; hence any state with $s(e) > 0$ for some finite-dimensional e — every normal state — fails. σ-completeness of the lattice is irrelevant; the obstruction is finitary. Only *singular* states ($s(e) = 0$ on all finite-dimensional e) escape, and whether one extends is open.

[3]The extension splits into two axes that the Boolean case fuses. The *extension axis* — passing from the

Even when probabilistic descent succeeds, value-definite descent does not. The Bub–Clifton theorem [2] addresses what *can* be salvaged:

Theorem 2.11 (Bub–Clifton [2]). *Given a state ρ on $L(H)$ and a preferred observable R, there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

(i) $D(\rho, R)$ is a Boolean algebra;

(ii) $D(\rho, R)$ contains the spectral projections of R;

(iii) The restriction of ρ to $D(\rho, R)$ is a mixture of dispersion-free states.

In the present vocabulary, condition (iii) says that $D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available — restating Halvorson and Clifton [11] in the EA/PR/VDR framework. It depends on R: different preferred observables yield different maximal Boolean subalgebras.

3 Distributivity as the commensurability condition

The status of the three positions depends on the algebra. In the OML case probabilistic realism splits into two grades that the Boolean case fuses: $\mathrm{PR}_{\mathrm{lat}}$, a σ-additive measure on the *lattice* A (Gleason, for $L(H)$), and $\mathrm{PR}_{\mathrm{dual}}$, a measure on the *dual* $S_0(A)$ concentrating on the physical points. We take $\mathrm{PR}_{\mathrm{dual}}$ to be the *stronger* grade — a dual-space measure restricts, on the $\perp$-stable clopens, to a finitely orthoadditive set function recovering s; promoting that to a σ-additive lattice measure ($\mathrm{PR}_{\mathrm{lat}}$) is free in the Boolean case but only conditional in general (see $(*)$). The relationship the two grades certainly exhibit is a *witnessed separation*, not a proven nesting: for $L(H)$, $\mathrm{PR}_{\mathrm{lat}}$ holds while $\mathrm{PR}_{\mathrm{dual}}$ fails. The positions, ordered with the stronger to the right (a node marked † is not realised for $L(H)$):

$$\underbrace{\mathrm{EA}}_{\perp\text{-stable set fn.}} \quad \Longleftarrow \quad \underbrace{\mathrm{PR}_{\mathrm{lat}}}_{\text{Gleason } (L(H))} \quad \overset{(*)}{\Longleftarrow} \quad \underbrace{\mathrm{PR}^{\dagger}_{\mathrm{dual}}}_{\text{fails for } L(H)} \quad \Longleftarrow \quad \underbrace{\mathrm{VDR}^{\dagger}}_{\text{KS-blocked (OML)}}$$

The implications run weak-from-strong: VDR (a global 2-valued homomorphism) implies $\mathrm{PR}_{\mathrm{dual}}$ (a dual-space measure); $\mathrm{PR}_{\mathrm{dual}}$ implies $\mathrm{PR}_{\mathrm{lat}}$ only conditionally (the restriction of $\hat{\mu}$ to the $\perp$-stable clopens is finitely orthoadditive, but its σ-additivity on the lattice needs $\hat{\mu}$ to be null on countable-join defects — free in the Boolean case, conditional in general; see $(*)$); and $\mathrm{PR}_{\mathrm{lat}}$ implies EA. For $L(H)$ the two strongest positions are not realised: VDR by Kochen–Specker, and $\mathrm{PR}_{\mathrm{dual}}$ by the clustering argument above — the *upward* extension construction $\mathrm{PR}_{\mathrm{lat}} \to \mathrm{PR}_{\mathrm{dual}}$ (build a dual-space measure from the lattice measure) is obstructed, governed by distributivity rather than σ-additivity. Gleason delivers $\mathrm{PR}_{\mathrm{lat}}$ but Gleason does not climb to $\mathrm{PR}_{\mathrm{dual}}$. The two grades coincide only in the Boolean case, where the extension is free and the ladder collapses to the single EA $\Leftarrow$ PR of Paper I. $(*)$ The arrow $\mathrm{PR}_{\mathrm{lat}} \Leftarrow \mathrm{PR}_{\mathrm{dual}}$ is free in the Boolean case but holds only conditionally in general (it requires a σ-continuity hypothesis on the dual measure that the finitary McDonald–Bimbó duality does not supply); we use it only in the direction stated. EA in the OML case is the $\perp$-stable clopen set function (the hedged reading below), not itself a dual-space measure.

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orthogonally additive state on the OML of $\perp$-stable clopens to a charge on the full Boolean algebra of clopens of $S_0(A)$ — is the classical Boolean extension problem [14]: since the representation map preserves meets, meet-zero elements map to disjoint sets, so the condition is $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families, a Pitowsky polytope condition [20] strictly stronger than orthoadditivity when the OML is non-distributive. It is not governed by σ-additivity, and in the finite case is a decidable linear program. The *descent axis* — whether the resulting measure concentrates on the physical points — is what σ-additivity governs. For $L(H)$, Gleason settles the lattice measure, but the extension axis itself fails (preceding footnote): Gleason does *not* deliver the dual-space measure.

Theorem 3.1 (Commensurability)**.** *Let A be a complemented lattice.*

(a) If A is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:

- *EA is unconditional (Stone construction).*
- *PR holds iff the charges are σ-additive* [18]*. Here PR_{lat} and PR_{dual} coincide: the extension step is free, so a lattice measure and a dual-space measure stand or fall together.*
- *VDR holds unconditionally (ultrafilters exist).*

Moreover, the choice of realisation space Ω is unconstrained by the algebra.

(b) If A is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:

- *EA is unconditional* in the $\perp$-stable set-function sense.[4]
- *PR_{lat} holds (Gleason, Theorem 2.8): a σ-additive measure on the lattice $L(H)$.*
- *PR_{dual}* fails *for every normal state: no charge on $S_0(L(H))$ exists (clustering argument above), so a fortiori none concentrating on $P(A)$. Equivalently, the extension step $PR_{lat} \to PR_{dual}$ is obstructed — the one transition the Boolean case leaves free.*
- *VDR fails globally (Kochen–Specker, Theorem 2.7).*
- *VDR holds locally within each Boolean context (Bub–Clifton, Theorem 2.11).*

Proof. Part (a) is Proposition 1.1. For part (b): $\mathrm{PR}_{\mathrm{lat}}$ follows from Gleason's theorem (Theorem 2.8), which supplies a σ-additive measure on the lattice $L(H)$ for any σ-additive state. $\mathrm{PR}_{\mathrm{dual}}$ fails for normal states by the clustering argument above: no charge on the clopens of $S_0(L(H))$ exists, so no Baire measure on $S_0(L(H))$ and a fortiori none concentrating on $P(A)$. EA holds in the $\perp$-stable set-function sense unconditionally — the assignment $\mu(h(a)) = s(a)$ on $\mathrm{CO}(S_0(A))^\dagger$ always exists; note this does *not* go via $\mathrm{PR}_{\mathrm{dual}}$ (which fails), since for $L(H)$ the genuine dual-space measure does not exist. VDR-failure is from Kochen–Specker; local VDR from Bub–Clifton. □

Corollary 3.2. *Distributivity is sufficient for VDR. For the physically central case $A = L(H)$ with $\dim H \geq 3$, non-distributivity blocks VDR.*

Proof. Sufficiency: Theorem 3.1(a). Blockage: Theorem 3.1(b). □

Remark 3.3. The dimensional restriction is necessary. For $\dim H = 2$, $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states. More generally, horizontal sums of Boolean algebras and certain finite OMLs are non-distributive yet admit dispersion-free states [16, Chapter 4] — though, as MO_2 shows, these dispersion-free states need not be 2-valued homomorphisms. The sharp characterisation for general OMLs is: VDR is available iff A admits a dispersion-free state, which is a property of A, not of distributivity alone. Distributivity is sufficient (ultrafilters exist) but not necessary in full generality. For the physically and epistemologically central case $\dim H \geq 3$, it is the effective dividing line.

The containment $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\mathrm{df}}(A)$ is standard in quantum logic [21, 16]. What we add is the identification of three transitions of different mathematical character, and the observation that distributivity controls which are trivial.

[4]EA here is the orthogonally additive set function $\mu(h(a)) = s(a)$ on the $\perp$-stable clopens, which always exists. It is *not* a genuine Baire measure on $S_0(A)$: promoting it to one is the extension step, which for $L(H)$ *fails* for every normal state (the clustering argument above) and for general OMLs is open. The unconditional claim is thus about the clopen set function, not the dual-space measure.

Definition 3.4 (Coherence filtration)**.** Let A be a complemented lattice with a family of charges. The *state space* of A admits a descending filtration:

$$\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\mathrm{df}}(A),$$

where:

- $\mathcal{S}(A)$ is the set of states on A (contextual coherence);
- $\mathcal{S}_\sigma(A) \subseteq \mathcal{S}(A)$ is the subset of σ-additive states (probabilistic coherence);
- $\mathcal{S}_{\mathrm{df}}(A) \subseteq \mathcal{S}_\sigma(A)$ is the subset of dispersion-free states (value-definite coherence).

Beneath the filtration sits a local condition: *consistency* requires only that each maximal Boolean subalgebra of A individually admits a compatible state.

Each inclusion can be strict. The three transitions have different mathematical character:

Transition	Character	Key	Obstruction
Consistency $\to \mathcal{S}(A)$	Pasting (topological)	Do local sections glue?	Contextuality
$\mathcal{S}(A) \to \mathcal{S}_\sigma(A)$	Regularity (analytic)	Does the state have good limits?	σ-additivity
$\mathcal{S}_\sigma(A) \to \mathcal{S}_{\mathrm{df}}(A)$	Sharpness (algebraic)	Can probabilities collapse to $\{0,1\}$?	Kochen–Specker

Distributivity collapses the filtration. For a Boolean algebra:

- *Pasting* is trivial: every local assignment extends to a global state (distributivity guarantees compatibility).
- *Sharpness* is free: ultrafilters exist (Zorn's lemma), so $\mathcal{S}_{\mathrm{df}}(A) \neq \varnothing$ unconditionally.
- The only non-trivial transition is *regularity*: σ-additivity of the charges. This is the content of [18].

For an OML with dim ≥ 3, every transition is non-trivial:

- *Pasting* is non-trivial: contextuality obstructions are possible.
- *Regularity* requires Gleason-type results. (This filtration grades *internal* states of A — i.e. $\mathrm{PR}_{\mathrm{lat}}$, the lattice measure, which Gleason supplies for $L(H)$; the further passage to a measure on the dual space $S_0(A)$, namely $\mathrm{PR}_{\mathrm{dual}}$ — the extension step, free in the Boolean case but blocked by non-distributivity — is a separate question that for $L(H)$ *fails* for normal states and for general OMLs is open.)
- *Sharpness* is blocked: $\mathcal{S}_{\mathrm{df}}(A) = \varnothing$ (Kochen–Specker).

Distributivity is sufficient to collapse the filtration. For $L(H)$ with dim ≥ 3, non-distributivity opens it fully.

Remark 3.5 (Relation to contextuality hierarchies)**.** Abramsky and Brandenburger [1] introduced a hierarchy of contextuality — probabilistic, possibilistic, and strong — grading the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our filtration is complementary: it grades *what the algebra can support*, not how badly a global section fails. Their axis measures obstruction strength; ours measures coherence demands.

4 Discussion

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above shows that the question is *parametrised* by the observation algebra. For Boolean observations, the mathematics does not prefer one position over another. For non-Boolean observations (dim ≥ 3), VDR is algebraically impossible; the constructive empiricist position [22] is the generic one, and value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

This reframing retroactively sharpens classical debates. The EPR argument for completeness [7] amounts to the demand that all observables simultaneously possess definite values — i.e., VDR. The Kochen–Specker theorem shows this is algebraically blocked for $\dim H \geq 3$. Bohr's complementarity corresponds to the non-triviality of the pasting transition: incompatible contexts cannot be simultaneously realised. Bell's beables [11] — the observables that *can* have definite values — are the elements of the maximal Boolean subalgebra $D(\rho, R)$ (Theorem 2.11). De Finetti's operationalism is EA in the Boolean case; Pitowsky [20] develops the quantum analogue via Dutch book arguments on $L(H)$.

The analysis is not specific to quantum mechanics. Any collection of individually Boolean observation contexts — jointly performable measurements in behavioural science [6], jointly queryable attributes in machine learning [4], simultaneously administrable treatments in experimental design — produces an observation algebra via categorical gluing [10]. If the contexts are mutually compatible, the colimit is Boolean and VDR is available. If not, the colimit is orthomodular and VDR may be blocked. The commensurability theorem thus provides a diagnostic across domains: when a deterministic latent-state model fails, the failure may be algebraic — the observation algebra does not support VDR — rather than a consequence of noise or insufficient data.

Several directions remain open:

1. *The OML extension problem.* Does every state on a general OML extend to a σ-additive measure on $S_0(A)$, as in the Boolean case? For $L(H)$ the answer is *negative* for *every* state: Gleason supplies the lattice measure but no dual-space charge exists — the clustering argument of §2 settles the normal states, and a finite family of infinite-dimensional, pairwise-meet-zero subspaces (a copy of MO_2 with infinite-dimensional atoms) settles the singular states. The general OML case remains open.

2. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti's theorem for exchangeability); the OML case is unexplored.

3. *Causal inference.* Holland's fundamental problem of causal inference [13] — that both potential outcomes $Y(0)$ and $Y(1)$ cannot be observed for the same unit — is structurally analogous to Bohr complementarity. Whether partial identification constraints generate non-Boolean algebraic structure, and whether the commensurability theorem applies to counterfactual reasoning, is unexplored.

4. *Relation to the topos programme.* The Bohrification programme [15, 5, 12] encodes related structure more powerfully: the spectral presheaf over the poset of Boolean subalgebras captures VDR-failure as absence of global sections (Kochen–Specker), and local VDR within each Boolean context is built into the presheaf's local sections. Our filtration operates at a coarser level — distinguishing EA, PR, and VDR as properties of the algebra's state space rather than as sheaf-cohomological obstructions. The McDonald–Bimbó duality offers a complementary perspective: a single topological dual space (not a

presheaf over contexts) in which principal filters are structurally distinguished. Whether the EA/PR/VDR vocabulary translates into the topos framework, and whether the filtration refines or merely simplifies the sheaf-theoretic picture, remains open.

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